

		$v = 5.55 \times 10^{-5} \text{ m s}^{-1}$
28	C	$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{1.3 \times 10^{-11}} = 5.1 \times 10^{-23} \text{ kg m s}^{-1}$ Gain in KE = Loss in electric potential energy $\frac{1}{2}mv^2 = e\Delta V$ $\Rightarrow \frac{1}{2} \frac{m^2 v^2}{m} = e\Delta V$ $\Rightarrow \frac{1}{2} \frac{p^2}{m} = e\Delta V$ $\therefore \frac{1}{2} \frac{(5.1 \times 10^{-23})^2}{9.11 \times 10^{-31}} = (1.6 \times 10^{-19})\Delta V$ $\Delta V = 8922 \approx 8.9 \text{ kV}$
29	D	Option A: The wavelengths of the characteristic peaks will change